

ONE RADEON. ONE ROBOT LOOP.

Radeon OneLoop

Single-Radeon Phase-Aware HIL Bimanual Handover

AMD Radeon Hackathon 2026 - Track 3: Physical AI

Team: Phi Media Lab

Formal profile: one AMD Radeon gfx1100 / ROCm 7.2.1

Project repository: github.com/phi-media-lab/radeon-oneloop

A reproducible bimanual handover pipeline that runs the Genesis environment, policy training, and real-time inference on a single Radeon while learning directly from reviewed human corrections.

Abstract

Radeon OneLoop is an auditable learning-and-deployment pipeline for a difficult SO-101 bimanual handover. The left arm must grasp and present a soft object, the right arm must receive it, and the system must place it in a target zone. Our main technical idea is phase-aware learning from human intervention: keep the architecture, optimizer and data fixed, but allocate more loss mass to the frames where a human corrected the policy and almost none to a failed autonomous prefix. The experiment compares this method against a uniform ACT baseline, initialized from scratch under an identical 10,000-step budget.

The competition boundary is intentionally strict. A single Radeon **gfx1100** executes the Genesis environment, both model-training jobs and policy inference. The CPU performs decoding, camera and robot I/O, validation, watchdog and emergency-stop duties only. Every accelerator job is tied to a GPU UID, immutable source commit, config hash, dataset hash, seed and raw evidence directory. The real dataset contains 124 episodes and 178,465 frames; 15,906 are reviewed intervention/correction frames. The formal dataset is train-only, so this report separates training-set diagnostics from task success. A reviewed 37/45 real-robot result is included only as historical evidence for the inherited closed-loop system, never as performance of the new formal checkpoints.

Under the predeclared step-10,000 rule, phase-aware ACT reduced normalized chunk L1 on sampled correction frames from 0.11957 to 0.11712 (2.05%), while the equal-role aggregate worsened from 0.09177 to 0.10479 (14.19%). This mixed train-frame diagnostic is consistent with reallocating capacity toward corrections, but it is neither validation nor task-success evidence. Full 100-action chunk median latency was 18.11 ms for baseline and 18.55 ms for phase-aware ACT on the same Radeon.

1. Problem and motivation

Bimanual handover combines perception, long-horizon coordination and contact. The receiving arm must arrive while the giving arm still stabilizes the object; release must occur only after a reliable grasp; and a small error early in the sequence can become an irrecoverable drop. These characteristics make the task a useful test of whether human interventions can be turned into targeted learning signal rather than simply appended to a dataset.

The physical platform consists of two six-command SO-101 arms and two RGB cameras. Each policy observation contains the 12 joint/gripper values plus a front and hand camera image. Each action contains five joint commands and one gripper command for the left arm followed by the same six commands for the right arm. The control contract is 30 Hz. ACT generates chunks of 100 actions, allowing the edge process to dispatch queued actions while the next chunk is computed.

The object and target vary less than they would in an unconstrained household setting. This is deliberate: the entry studies reliable phase transitions and intervention learning in a complete real pipeline, rather than claiming general-purpose manipulation.

2. Compliance and system architecture

2.1 One-card boundary

The formal host exposes one ROCm agent and one PyTorch device:

Item	Formal value
ROCm / HIP	7.2.1 / 7.2.53211
PyTorch	2.9.1+rocm7.2.1
Genesis	1.3.1, gs.amdgpu
Target	gfx1100
Reported name	AMD Radeon Graphics
GPU UID	0x153f7d55778ab659
VRAM	51,522,830,336 bytes (47.98 GiB)
Visible accelerator count	1

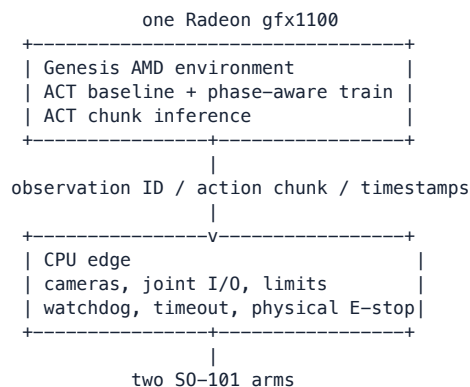
PyTorch uses its CUDA-compatible Python namespace for ROCm devices. Therefore Genesis/PyTorch logs can display **cuda:0** even though the underlying runtime is HIP. The simultaneous **torch.version.hip**, **gfx1100**, AMD device name, ROCm UID and dependency audit establish that this is the Radeon device; the bootstrap rejects NVIDIA packages and no CUDA runtime is present.

The same device executes:

- 1 Genesis scene build, rigid-body stepping and camera rendering;
- 2 ACT baseline training;
- 3 phase-aware ACT training; and
- 4 full-chunk ACT inference and queued-action dispatch.

radeon-f is an equivalent but distinct shadow GPU used to fail fast on clean environment and two-step smoke tests. A Radeon APU machine and an MI300X machine were inspected for prior work only. Their checkpoints, frame rates and training results are forbidden formal sources.

2.2 Runtime split



The CPU is not an alternative inference path. The formal profile explicitly prohibits CUDA, an NPU, a second GPU, a software renderer and remote inference APIs.

2.3 Evidence discipline

Every GPU command runs under an exclusive file lock. The launcher asserts one PyTorch device, a **gfx1100** architecture and the expected formal UID. It then writes the exact command, environment freeze, hardware record, configuration, one-second ROCm samples, stdout/stderr, metrics, hashes and **DONE** or **FAILED**. Failed and aborted runs are retained. Only registered runs may populate result tables.

3. Data and contracts

3.1 Sources

The immutable dataset builder merges two team-collected sources without mutating either:

Source	Episodes	Frames	Role
Human behavior demonstrations	84	116,550	Train
Reviewed HIL policy rollouts	40	61,915	Train
Combined	124	178,465	Train only

The builder verifies two $480 \times 640 \times 3$ RGB streams, the 30 Hz rate, the 12-value action and state order, task tables, video references, contiguous episode indices and exact HIL manifest coverage. It emits a file-level SHA-256 ledger and refuses to overwrite a prior output. The resulting dataset hash is:

```
ba18dd207ffd00c562a7ad18c831508d0529cd4d8d7b478a9b2f6d46618489cf
```

The raw dataset is access-controlled because it contains team-recorded robot video and has not been cleared for public redistribution. The schema, builder, source hashes and derived hashes are public.

3.2 Action and observation contract

The frozen action order is:

```
left:  shoulder_pan, shoulder_lift, elbow_flex, wrist_flex, wrist_roll, gripper
right: shoulder_pan, shoulder_lift, elbow_flex, wrist_flex, wrist_roll, gripper
```

Real revolute joints are expressed in degrees, while the gripper uses closed = 0 and open = 100. The Genesis adapter converts revolute joints to radians and maps the gripper monotonically into the model joint range. Round-trip and ordering tests run without robotics or ML dependencies.

3.3 Intervention labels

The 40 HIL episodes were human-reviewed. Successful policy frames receive the ordinary behavior weight. In failed episodes, explicit correction intervals identify where the operator took over; the preceding autonomous segment is labeled as a failed policy prefix.

Role	Frames	Raw weight	Normalized gradient-mass share
Behavior demonstration	116,550	1.00	54.24%
Successful policy	34,101	1.00	15.87%
Failed policy prefix	11,908	0.05	0.28%
Human correction	15,906	4.00	29.61%

Where phase-aware ACT spends its gradient

Correction frames are 8.91% of data but receive 29.61% of normalized gradient mass

Share of training frames



Share of gradient mass



Frame share versus normalized gradient-mass share in the frozen phase targets.

Weights are normalized over positive frames so the positive mean is one. This keeps the overall loss scale comparable while changing which frames dominate the gradient. The sidecar hash is:

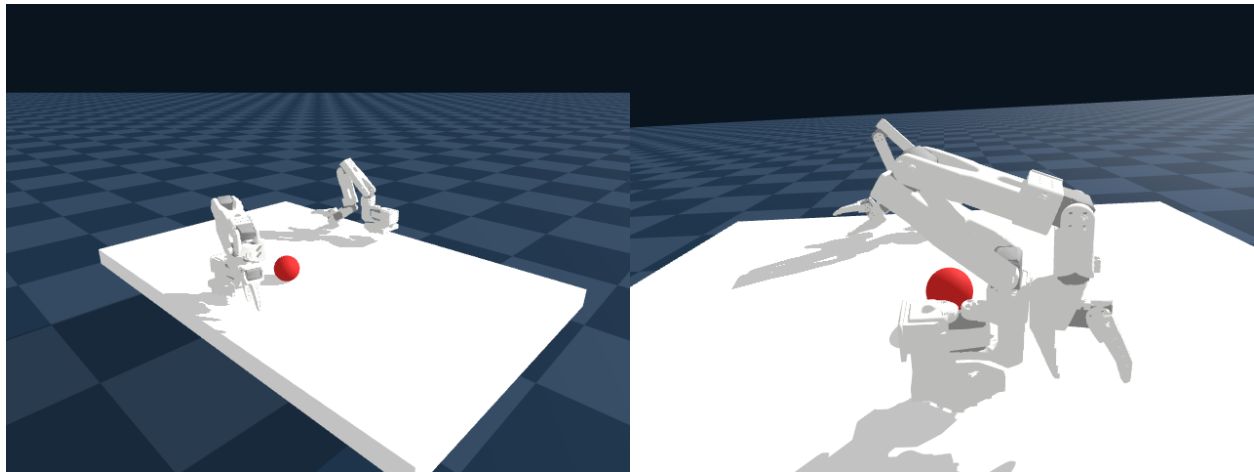
b34f5559db6d697890a08b6a7e49b52f3a916537c3328aea72c8470bb90bec57

4. Genesis environment

The simulation component is deliberately minimal. It downloads and verifies the official SO-101 MJCF and mesh assets, instantiates two arms, a table, a rigid object and two cameras, and exposes the same state and image keys as the real dataset. It includes reset, position control, deterministic stepping, attached-camera motion and an explicit placement predicate.

The camera-corrected formal smoke ran 1,000 steps on `gs.amdgpu`, exercised both arms, rendered the front and attached hand camera, and validated finite 12-value state. Scene build took 33.84 s. The recorded all-step latency, including the capture steps, was:

Statistic	Time
Median	4.27 ms
p95	5.02 ms
p99	7.02 ms



Corrected formal Genesis front camera (left) and attached hand camera (right).

The scripted motion is a build/control/camera test and does not execute a learned handover; `task_success=false` is recorded in the formal metrics. The earlier formal scene run remains public but its hand-camera image is excluded from visual claims after the relative camera transform was corrected.

5. Policy method

5.1 Baseline

The baseline is the default LeRobot ACT architecture (51.6 million parameters in the smoke evidence), initialized randomly and trained on every frame with uniform sample contribution. Its role is a reproducible control, not a historical checkpoint.

5.2 Phase-aware ACT

The phase-aware model uses the same architecture, random initialization, observations, action targets, optimizer defaults, batch size, worker count, seed and 10,000-step budget. Only the per-frame ACT loss weighting changes. The patched training path looks up each sampled dataset index in the immutable sidecar, applies the weight to the per-item loss and normalizes nonzero batch weights. A startup audit reports coverage and weight statistics before the first optimizer update.

The design encodes a simple hypothesis: the short human correction after a failure contains more information about recovery and transition timing than a long autonomous prefix already known to be unsuccessful. Weighting avoids duplicating frames and preserves an exact baseline comparison.

5.3 Selection rule

The dataset has no held-out validation split, and the Genesis scene is not a calibrated replica suitable for choosing a real-task checkpoint. We therefore predeclared the final checkpoint at step 10,000 as the only candidate for each method. Checkpoints saved every 1,000 steps remain debugging artifacts and are not searched after observing loss.

5.4 Key technical contributions

- a deterministic frame-level intervention objective that plugs into ACT without changing its architecture or duplicating training samples;
- a matched real/sim observation and action contract for two SO-101 arms, including a moving hand-camera transform verified on `gs.amdgpu`;
- a fail-closed single-Radeon job protocol that makes hardware identity, source, data, configuration and final model content independently auditable; and
- a CPU-edge chunk protocol whose timeout, sequence, range and delta checks latch E-stop before unsafe values reach robot I/O.

6. CPU-edge safety

The dependency-free safety controller uses monotonically increasing sequence IDs and monotonic-clock timestamps. Before any chunk can reach robot I/O it requires:

- an armed, non-latched controller;
- a current observation and command;
- exact observation/action sequence correspondence;
- a non-empty, finite 12-DoF action chunk;
- joint and gripper values inside configured limits; and
- per-joint delta below the configured bound.

A stale observation, stale command, reordered packet, mismatch or limit violation latches E-stop. Software recovery is not permitted: a new controller must be created after physical reset. These checks supplement rather than replace manufacturer safety limits, workspace clearance and an operator with access to the physical stop.

7. Evaluation protocol

7.1 Training and runtime measurements

Both formal runs use the same machine, dataset hash, seed, batch size and training budget. We report elapsed wall time, terminal training loss, checkpoint hash and externally sampled peak GPU memory. These are systems and optimization measurements, not robot-success estimates.

For inference, a real dataset observation is decoded into the same camera and state dictionary used at runtime. After 20 warm-up calls, 200 synchronized full-chunk calls measure model execution plus preprocessing/postprocessing. The policy is then invoked without reset to measure queued action dispatch. The report includes mean, median, p95 and p99 latency, action shape, finiteness, chunk horizon and peak allocated VRAM.

7.2 Action reconstruction diagnostic

The evaluator selects up to 256 evenly spaced frame indices from each segment role. It compares the predicted normalized ACT action chunk and the first action against dataset targets. Selection is deterministic and identical for both checkpoints. Because all samples are training frames, the values are reported as reconstruction diagnostics only. They cannot estimate generalization or closed-loop task success.

7.3 Real-robot boundary

The source project contains a complete physical perception-policy-control loop. A prior reviewed 45-episode batch recorded 37 successes and eight **handover_failed** outcomes (82.22%). Hashes of the run, annotations, action log, launch ledger and checkpoint were captured from a read-only evidence host. That checkpoint predates this formal experiment and is neither a parent nor a result. No success rate for the new checkpoints is claimed without new, supervised physical rollouts.

8. Formal results

8.1 Environment

Run	Result	Formal evidence
ROCM/PyTorch/Genesis clean bootstrap	PASS	one gfx1100 , AMD matmul, gs.amdgpu , Vulkan enumeration
Dataset build	PASS	124 episodes, 178,465 frames, exact derived hash
Genesis dual-arm smoke	PASS	1,000 steps, two cameras, 12-value state

8.2 Paired ACT experiment

Metric	Baseline ACT	Phase-aware ACT
Updates	10,000	10,000
Training wall time	87.75 min	87.68 min
Terminal logged training loss	0.113	0.134
Mean sampled GPU utilization	98.59%	98.45%
Peak sampled VRAM allocation	17% (≤ 8.16 GiB)	17% (≤ 8.16 GiB)
Step-10,000 checkpoint SHA-256	7c8f2089...29dc79	3ae18054...2721d4

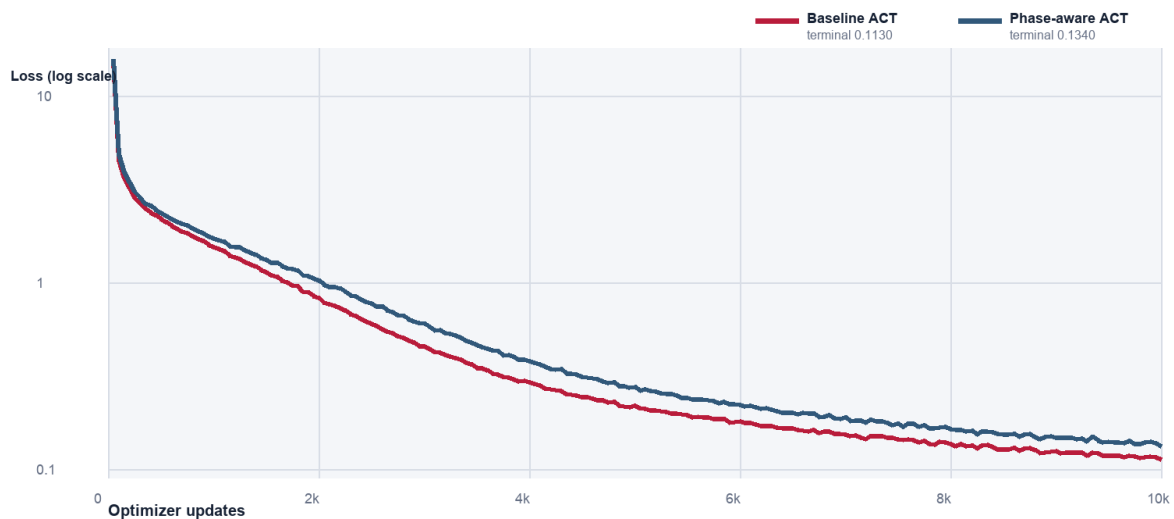
The complete artifact-tree digests are:

```
baseline    7c8f2089c2f9ff5632ab1272754bdece46e30280ffce6c8ecde850956429dc79
phase-aware 3ae1805441889adf3fcaa23ff45e79509f0be747619c2783227014bc162721d4
```

The logged losses are optimized under different frame-weight distributions. They document convergence but are not an accuracy ranking.

Formal ACT training loss

Same Radeon, dataset, architecture, seed, batch size, optimizer, and 10,000-step budget



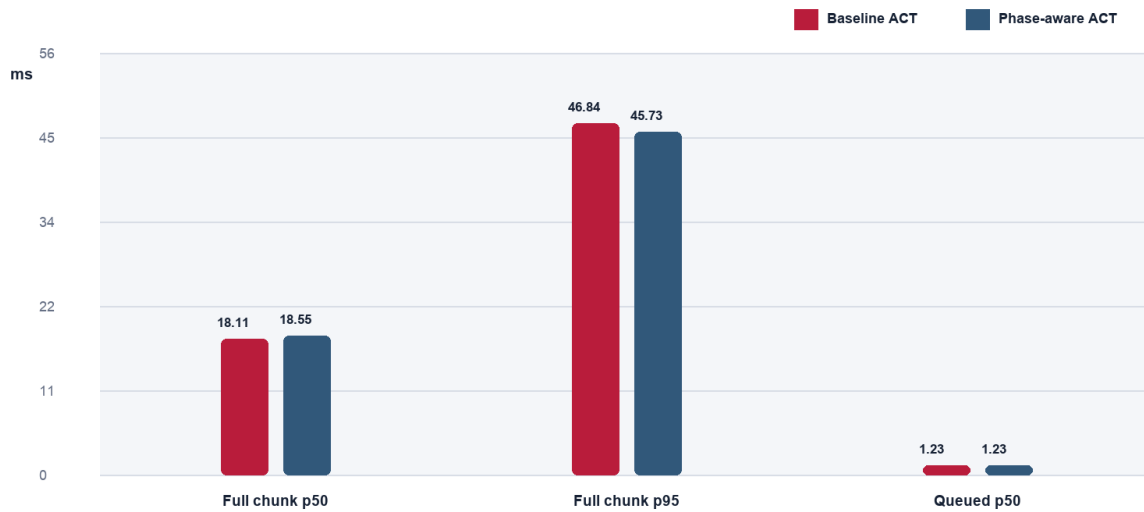
Matched formal ACT training loss on the single Radeon.

8.3 Inference and reconstruction

Metric	Baseline ACT	Phase-aware ACT
Full 100-action chunk, median	18.11 ms	18.55 ms
Full chunk, p95	46.84 ms	45.73 ms
Queued action dispatch, median	1.233 ms	1.233 ms
Peak allocated inference VRAM	377.8 MiB	377.8 MiB
Equal-role normalized chunk L1	0.09177	0.10479
Correction normalized chunk L1	0.11957	0.11712
Correction normalized first-action L1	0.09009	0.09626

Single-Radeon ACT inference latency

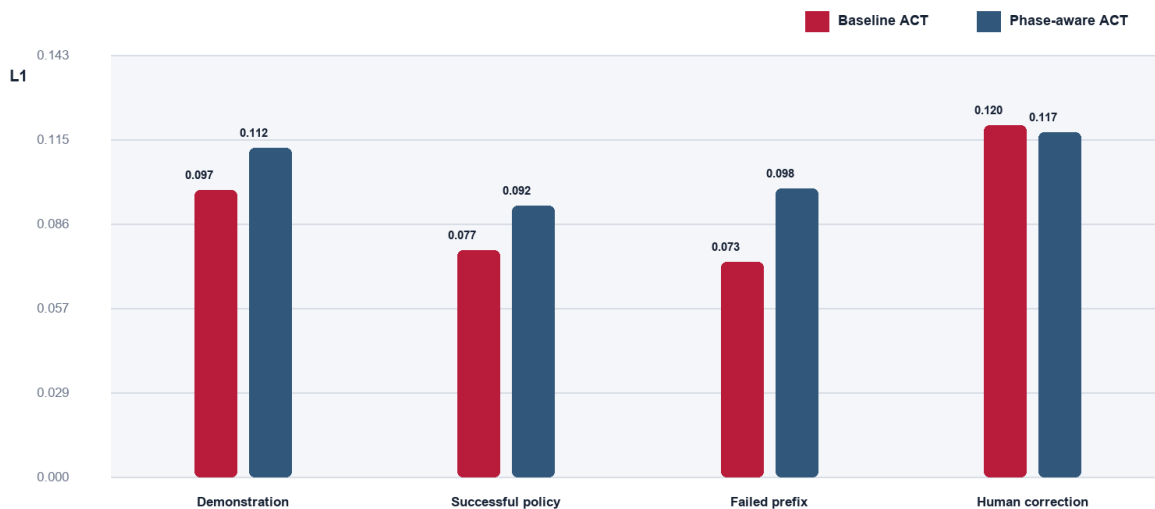
200 synchronized full-chunk calls after warm-up; real dataset observation; 100-action horizon



Synchronized full-chunk and queued-action latency on the single Radeon.

Train-frame action reconstruction by segment role

Deterministic equal-role sampling; normalized chunk L1 is a diagnostic, not task success



Role-stratified train-frame reconstruction; lower is better, but this is not a task-success metric.

8.4 Interpretation

The phase-aware checkpoint improved the metric it explicitly emphasized: correction-frame chunk L1 decreased 2.05%. The cost is visible rather than hidden: the equal-role aggregate worsened 14.19%, demonstration L1 rose from 0.09745 to 0.11187, successful-policy L1 from 0.07705 to 0.09213, and failed- prefix L1 from 0.07301 to 0.09805. Correction first-action L1 also worsened 6.85%, so the small correction gain is distributed over the predicted chunk, not its immediate action. This supports a capacity-reallocation interpretation but does not establish better closed-loop recovery. No table value comes from a shadow host, smoke checkpoint, MI300X, APU/NPU, post-hoc checkpoint choice or unregistered directory.

9. Negative results and scope decisions

9.1 Gaussian workspace twin

We evaluated adding a Vulkan/RADV Gaussian workspace representation through VkSplat. Existing machines contained a corgi example, synthetic cameras and unrelated 3D/4D Gaussian research, but not a calibrated static multi-view capture of the competition SO-101 workspace. Substituting those assets would make an attractive demo without demonstrating the same task. The branch was therefore gated out before formal optimization. Its hash-pinned runner and capture schema remain as reproducible future work, but Gaussian output is not an observation, result or claimed contribution in this release.

9.2 Aborted checkpoint-selection run

An initial formal baseline was stopped at step 250 after discovering that its configuration text described fixed validation seeds even though the dataset has no validation split. The run remains marked failed. Before restarting, the selection rule was changed to the predeclared final training step and covered by a regression test. Preserving this negative run is part of the evidence policy.

9.3 General limitations

- The formal dataset is train-only and cannot quantify generalization.
- The Genesis scene verifies AMD execution and interfaces but is not a calibrated sim-to-real benchmark.
- Historical physical success does not measure either new checkpoint.
- Raw team video cannot be redistributed, limiting one-command external data reproduction; authorized reviewers can reproduce from registered hashes.
- The task is constrained to one object family and workspace.
- The CPU-edge safety kernel has unit tests but no safety certification.

10. Reproducibility and final deliverables

The final output forms are this English PDF technical report, the dedicated Apache-2.0 source repository and detailed root README, a public 3-5 minute H.264/AAC workflow video, machine-readable formal evidence, and the official competition pull request. The public repository contains:

- exact environment and dependency bootstraps with wheel/source hashes;
- immutable data merger, phase-target builder and public source/derived hashes;
- frozen baseline, phase-aware and single-Radeon configurations;
- a hash-verified SO-101 Genesis asset fetcher and minimal scene;
- fair-pair checks and exact LeRobot command generation;
- latency, reconstruction and safety evaluators;
- job manifests, GPU sampling, terminal markers and SHA-256 ledgers; and
- unit and shell validation invoked by `./ops/validate_scaffold.sh`.

The root README gives clean-clone commands and makes the access-controlled data boundary explicit. Project code is Apache-2.0; dependency and asset status is listed in `THIRD_PARTY_NOTICES.md`.

11. Upstream community contribution

During the competition we filed [Genesis issue #3163](#) to improve the project's AMD support path. The checked-in [docker/Dockerfile.amdgpu](#) still describes a ROCm 6.4.1 / PyTorch 2.6 / Genesis 0.2.1-era target, while Radeon OneLoop provides public native-host evidence for ROCm 7.2.1, AMD PyTorch 2.9.1 and Genesis 1.3.1 on [gfx1100](#). The report proposes an explicit supported tuple, pinned compatibility-sensitive packages and a [gs.amdgpu](#) scene smoke. It also states that we did not build the current Dockerfile, avoiding an unverified container claim. A code PR is offered once maintainers select the intended base-image tag.

12. Team contribution

Baoshu Feng, Phi Media Lab - project and experiment design; SO-101 data and HIL workflow integration; intervention-target methodology; Radeon/ROCm bring-up; Genesis environment; formal training and evaluation; CPU-edge safety; evidence, report, and submission engineering.

The underlying robot collection and reviewed HIL artifacts were produced in Phi Media Lab's prior real-robot workflow. Upstream Genesis, PyTorch, ROCm, LeRobot and SO-101 assets remain credited to their respective authors.

References

- 1 AMD-DEV-CONTEST, [Radeon Hackathon 2026-07 official repository](#).
- 2 Genesis Authors, [Genesis source and documentation](#), version 1.3.1.
- 3 Zhao et al., [Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware](#), ACT / ALOHA.
- 4 Hugging Face, [LeRobot](#).
- 5 AMD, [ROCm Radeon documentation](#) and official PyTorch wheels.